5.6. Representations of products

Theorem 5.6.1. *Let G, H be finite groups, let $\{V_i\}$ be the irreducible representations of G over a field k (of any characteristic), and let $\{W_j\}$ be the irreducible representations of H over k. Then the irreducible representations of $G \times H$ over k are $\{V_i \otimes W_j\}$.*

Proof. This follows from Theorem 3.10.2. □

5.7. Virtual representations

Definition 5.7.1. A **virtual representation** of a finite group G is an integer linear combination of irreducible representations of G, $V = \sum n_i V_i$, $n_i \in \mathbb{Z}$ (i.e., n_i are not assumed to be nonnegative). The character of V is $\chi_V := \sum n_i \chi_{V_i}$.

The following lemma is often very useful (and will be used several times below).

Lemma 5.7.2. *Let V be a virtual representation with character χ_V. If $(\chi_V, \chi_V) = 1$ and $\chi_V(1) > 0$, then χ_V is a character of an irreducible representation of G.*

Proof. Let $V_1, V_2, \ldots, V_m$ be the irreducible representations of G, and let $V = \sum n_i V_i$. Then by orthonormality of characters, $(\chi_V, \chi_V) = \sum_i n_i^2$. So $\sum_i n_i^2 = 1$, meaning that $n_i = \pm 1$ for exactly one i and $n_j = 0$ for $j \neq i$. But $\chi_V(1) > 0$, so $n_i = +1$ and we are done. □

5.8. Induced representations

Given a representation V of a group G and a subgroup $H \subset G$, there is a natural way to construct a representation of H. The **restriction** of V to H, $\mathrm{Res}_H^G V$ is the representation given by the vector space V, and the action $\rho_{\mathrm{Res}_H^G V} = \rho_V|_H$.

There is also a natural, but less trivial, way to construct a representation of a group G given a representation V of its subgroup H.